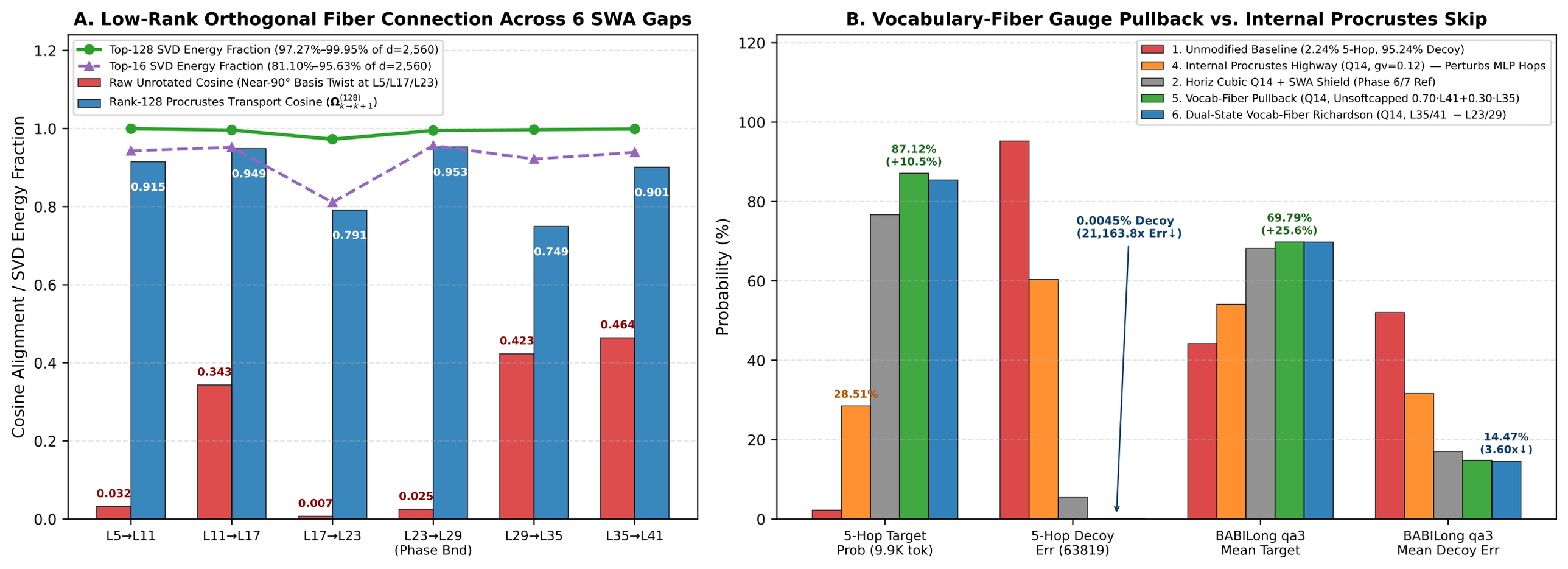
A. Low-Rank Orthogonal Fiber Connection Across 6 SWA Gaps
Top-128 SVD Energy Fraction (97.27%–99.95% of d=2,560)
Top-16 SVD Energy Fraction (81.10%–95.63% of d=2,560)
Raw Unrotated Cosine (Near-90° Basis Twist at L5/L17/L23)
Rank-128 Procrustes Transport Cosine ($\Omega^{(128)}_{k\to k+1}$)
Cosine Alignment / SVD Energy Fraction
0.032
0.915
0.343
0.949
0.007
0.791
0.025
0.953
0.423
0.749
0.464
0.901
L5→L11
L11→L17
L17→L23
L23→L29
(Phase Bnd)
L29→L35
L35→L41
B. Vocabulary-Fiber Gauge Pullback vs. Internal Procrustes Skip
1. Unmodified Baseline (2.24% 5-Hop, 95.24% Decoy)
4. Internal Procrustes Highway (Q14, gv=0.12) — Perturbs MLP Hops
2. Horiz Cubic Q14 + SWA Shield (Phase 6/7 Ref)
5. Vocab-Fiber Pullback (Q14, Unsoftcapped 0.70·L41+0.30·L35)
6. Dual-State Vocab-Fiber Richardson (Q14, L35/41 − L23/29)
Probability (%)
28.51%
87.12%
(+10.5%)
0.0045% Decoy
(21,163.8x Err↓)
69.79%
(+25.6%)
14.47%
(3.60x↓)
5-Hop Target
Prob (9.9K tok)
5-Hop Decoy
Err (63819)
BABILong qa3
Mean Target
BABILong qa3
Mean Decoy Err